

ENCN442 Coastal Engineering Design Project

Due date: 9am Monday 11th May 2026

Expected time: 3.5 hours

Project Description and Scope

This project will require the third and fourth modules of the Coastal Engineering section of the course (Wave Transformations and Wave Design).

You have been asked to model the wave field where a new pier is being constructed. To help with this, a high frequency wave record has been provided, but it is sourced from further offshore, so will need to be modified for the design site.

The wave record that you're given contains a large number of waves. You've also been told that you might get a longer or different wave record at a later stage in the project (not in this assignment), so it will be best to analyse the wave record using Python, rather than manually or with Excel. If you need help with this, I'm more than happy to help!

I'm also happy for you to use a large language model, like Claude, to assist you in analysing the wave record. However, if you do so, it is incredibly important that you validate the result from any piece of code to make sure that it is doing what you'd expect it to. Also be very careful installing any packages that you don't recognise.

You should present your results in a way that makes it clear what you have done at each step. You don't need to describe every calculation but a broad understanding of your process is important.

Project

A wave record is available on Learn which gives the measured wave height every 0.25 s for 60 minutes. The measurement location was significantly offshore from the design site where the water depth is 17 m. At the wave record location, waves typically approach the coast at an angle of 25° (0° waves would have crests that were parallel with the coastline). It has been decided that the $H_{1/100}$ wave based on the short term wave record will be used for the design.

1. Use an up-crossing analysis to calculate $H_{1/100}$ and T_z , the mean period between zero upward crossings. Also give the significant wave height, H_s that is needed to calculate $H_{1/100}$. Hint: The numpy function *sign* can be very helpful for your up-crossing analysis.

(15 marks)

You know that the properties of waves can change as they move into shallower water. The water depth where the pier is being constructed is 0.9 m at mean tide. However, you're not sure if the waves at either the measurement or design location should be categorized as shallow, deep, or transitional waves. The group

velocity for a transitional wave is given by:

$$c_g = \frac{1}{2} \sqrt{\frac{g \tanh(kh)}{k}} \left(1 + \frac{2kh}{\sinh(2kh)} \right).$$

2. Classify the waves at the design and measurement locations as shallow, deep, or transitional. Use this classification to calculate the design parameters ($H_{1/100}$, T_z and the angle that the waves will make with the coastline) at the design site.

(10 marks)

The pier will be supported by cylinders with a diameter of 0.3 m. You know that the Morrison equation should be used to predict the force per unit length for these supports and think that suitable values for the constants are $C_D = 0.9$ and $C_I = 2$.

3. Plot the drag, inertial, and total force per unit length acting on the pier supports for one full wave period. Calculate the maximum force per unit length over the entire wave phase from this plot.

(5 marks)

4. Approximately how many hours did you spend working on this assignment?

(0 marks)